

cations in military, law enforcement, surveillance, security, navigation, hunting, wild life observation and so on. These devices are used not only by the military, police forces and law enforcement agencies for navigation, surveillance and targeting, these are also used by private detectives, fire-fighters, hunters and nature enthusiasts. Both image enhancement and thermal imaging night vision equipment have preferred areas of application to which these are better suited than the other.

Image enhancement night vision devices. Image intensification night vision needs at least a small amount of light to operate. This ambient light, in some cases, may be provided by the moon and stars. In the case of total darkness, the night vision device can see no better than the naked eye. This shortcoming is overcome with the use of an IR illuminator, which works like a flashlight for night vision while staying invisible to the naked eye.

Use of an IR illuminator has a drawback. It makes the device highly prone to be detected by enemy's night vision device.

Night vision technology is used extensively by the military and police due to the instant tactical advantage they get through the ability to see in the dark, and precisely detect and even identify the target even from a distance. AN/PVS-14 (Fig. 7) is one such versatile night vision scope extensively used by armed forces and special operations units. It is a monocular night vision device designed for use by the individual soldier in a variety of ground-based night operations. While the dark-adapted unaided eye provides situational awareness and vision of close-range objects, the night vision device provides long-range vision of potential threats and targets, thus, making it a force multiplier in night time warfare.

Night vision cameras are extensively used for surveillance and security applications, especially for around-the-clock surveillance, indoors and in controlled environments. Such cameras provide enhanced security by improving video images in low-light conditions. With the help of IR illumination, these can be effectively used in near total dark conditions, too. Per-



Fig. 7: AN/PVS-14 night vision scope



Fig. 8: FLIR HRC-E

haps, the greatest feature of night vision cameras is that these are true 24x7 cameras, as these are the only cameras with the ability to record any event in a protected area around the clock when connected to a digital video recorder.

Night vision cameras are available in two basic styles: dome and bullet. The resolution typically varies from 400 TV lines to 700 TV lines. Security cameras with long-range night vision find extensive usage in monitoring large parking lots, huge and dark warehouses, apartment complexes and other similar assets on 24x7 basis. Traffic monitoring and monitoring of suspicious activity in public places, security installations and critical national assets are other important application areas of security cameras with night vision capability.

Thermal imaging night vision devices. Thermal imaging devices are used on land-based armoured fighting vehicles, naval vessels and aerial platforms such as aircraft, helicopters and missiles for surveillance, target acquisition and tracking applications. Some common non-military applications of these systems include surveillance of living things, search and rescue operations during fire-fighting, detection of gas leaks, monitoring of volcanoes and detection of heat in faulty electrical joints.

In military applications, thermal imaging devices offer several distinct advantages. The first is their immunity to detection by the adversary. These are passive sensors that do not emit any radiation for generation of images.

Second, it is extremely hard to camouflage the target from the sensor as these sense heat radiation.

Third, thermal imagers can see through smoke, fog, haze and other atmospheric obscurants better than sensors operating in the visible spectrum.

One limitation of thermal imaging sensors is that these cannot discriminate between a friend and a foe.

Friendly forces can use heat beacons to overcome this shortcoming.

Thermal imagers can be extremely useful in fire-fighting operations. Their ability to see through smoke and debris allows fire-fighters to find people who have passed out and those fighting for survival but are too afraid to come out from hiding. Thermal imagers can also tell a fire-fighter if a door is hot and possibly contains a fierce blaze on the other side.

Thermal imagers designed for fire-fighting missions are designed to operate in very rough and tough environmental conditions.

Thermal imagers are extensively used by law enforcement agencies and armed forces for a variety of applications in navigation, detection and targeting. These allow them to detect potential threats without exposing their location to the adversary. State-of-the-art thermal imaging rifle scopes have become rugged enough to withstand the abuse of recoil, making these extremely popular with the armed forces.

While thermal imagers are incredibly effective when it comes to detecting human beings or animals, discrimination between a friend and a foe may be a challenge. This may be an issue in life-and-death situations.

Thermal imaging cameras are one of the most effective tools for surveillance because these work equally well during the day and at night. Regular CCTV cameras are limited by their need for light, and night vision devices do not function during the day.

The chance to see through smoke and fog also gives thermal a leg up on other surveillance techniques. Fig. 8 shows a thermal imaging sensor designed for border and coastal surveillance array. It offers continuous zoom from 25° to 2°, and a long-range detection capability. The sensor can detect a human being-sized target from more than 15 kilometres away.

State-of-the-art sensors intended for surveillance, target detection and tracking applications employ multi-sensor configurations, often combining a visible spectrum sensor with an IR sensor. Final image in this case is the result of fusion of imaging data from two sensors.

To be continued